

# Yeqiu (Jason) He

AI Systems Engineer — Agentic Systems · RAG · LLM Integration

Australian Permanent Resident — Subclass 190 (Skilled Nominated, granted). Full work rights, no sponsorship required.

Relocating to Australia (currently Stuttgart, Germany) · Available immediately

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**Proof of work** — every project below is open-source on GitHub, with each build documented on a technical blog and YouTube; Meinrag is in active production use.

## PROFESSIONAL SUMMARY

AI Systems Engineer who designs, ships and operates production LLM systems end-to-end — multi-agent orchestration, retrieval-augmented generation (RAG) and Model Context Protocol (MCP) — owning the full lifecycle from architecture through deployment to live operation on self-hosted infrastructure. Grew into AI engineering from 5+ years shipping enterprise process-automation that cut manual processing by 70–90% across 30+ enterprise systems, then rebuilt that same systems-engineering and evaluation discipline around LLMs. Physics research background (2 first-author papers) drives a rigorous focus on evaluation, calibration and reproducibility. Seeking an AI Engineer / AI Integration Engineer role in Australia applying hands-on RAG, MCP and ML-pipeline depth to real-world production systems.

## SKILLS

**AI / ML:** LLM integration & orchestration (OpenAI, Claude, Codex/GPT) · Retrieval-Augmented Generation (FAISS, embeddings, re-ranking, hybrid BM25/dense retrieval) · MCP (Model Context Protocol) servers · agentic / multi-agent systems · prompt engineering · CatBoost / ML · model calibration & evaluation · NLP · vector search · Monte Carlo simulation

**Programming:** Python · VB.NET · C# · VBA · JavaScript · TypeScript · SQL · Fortran

**Backend / Data:** FastAPI · SQLAlchemy · PostgreSQL · MySQL · SQLite · data-pipeline / ETL design · REST APIs

**Automation / RPA:** UiPath · PowerShell · Windows automation · process optimisation

**Infrastructure / Tools:** Docker · Linux · Git · Raspberry Pi / NAS / VPS · Visual Studio · VS Code · Cursor

**Frontend:** Vue.js · React · Nuxt · Tailwind

**Languages:** English (PTE 65 ≈ IELTS 7.0, full professional proficiency) · Mandarin Chinese (native) · German (A2, basic)

## EXPERIENCE

### AI Systems Engineer — Independent (open-source · in production)

Jan 2026 – Present

Stuttgart, Germany (relocating to Australia)

Design production AI systems — multi-agent orchestration, RAG and ML pipelines — and direct AI agents to execute the build work, owning architecture, integration, deployment and operation on self-hosted infrastructure (Raspberry Pi 5 + Synology NAS + VPS).

- **OpenClaw — production multi-agent LLM system:** designed and run a fleet of 8 specialised agents (general assistant, research, engineering, maths, media, review, legal, supervisor) in continuous production, coordinated over a custom inter-agent message-bus protocol. Engineered a persistent anchor-based memory with semantic + hybrid (cosine/BM25) retrieval for cross-session context recall, scheduled autonomous tasks, and multi-provider LLM routing (Claude / OpenAI / GPT-Codex) with automatic fallback for provider resilience.
- **Meinrag — multimodal RAG platform (open-source, in production):** built end-to-end and now serving a legal corpus of 3,300+ documents — hybrid retrieval (BM25 + dense, RRF) with LLM re-ranking returns PDF bounding-box citations that keep every claim traceable to source. Stack: PostgreSQL + FAISS/Chroma vector store + FastAPI + React, Docling document ingestion, exposed via an MCP server with Bearer auth and audit logging; Dockerised and test-covered.
- **Cicero — AI legal-research assistant (in production):** built on Meinrag in collaboration with a practising lawyer — automates retrieval-augmented legal-memo drafting and an end-to-end email → retrieval → document → email pipeline with PII-boundary handling.
- **ML prediction pipelines:** built CatBoost + Elo/pi-rating football models (~36 engineered features, per-league calibration) and a 50,000-run Monte Carlo tournament simulation. Rigorous out-of-sample evaluation — rolling-origin backtesting, bootstrap confidence intervals and pre-registered ("Ulysses-pact") decision rules to prevent overfitting; a daily automated digest tracks model-vs-market calibration (RPS).

- **MCP servers** (time-mcp, meinrag-mcp) exposing scheduling/time tools and the Meinrag corpus to LLM agents; **content automation** — Ghost CMS publishing via Admin API (JWT, KaTeX) and a TTS voice-cloning video pipeline (ffmpeg + Qwen3-TTS).

#### IT Supervisor — SGS Standard Technical Services Co., Ltd.

Jan 2021 – Jan 2026

Shanghai, China / Stuttgart, Germany

Lead developer for end-to-end process automation and AI integration across departments at SGS (global testing, inspection & certification); owned the full lifecycle from requirements to deployment, maintenance and user training.

- Developed and deployed **30+ automation systems** (document generation, data synchronisation, process monitoring) using Python / FastAPI backends, .NET components and UiPath orchestration; cut manual processing time by **70–90%** in key workflows.
- Introduced **AI-enhanced workflows** — embeddings, similarity search and OpenAI GPT integration — for report validation, content comparison and document intelligence (the foundation for the AI systems work above).
- **AI-assisted Report Automation** (Python, VB.NET, SQLite, UiPath): extracts photos and data into Word templates with auto-dispatch — **~70 reports/month, ~1 hour → minutes**.
- **Certificate Upload & Compliance Automation** (VB.NET, Python, UiPath): parses emails, extracts Word/PDF metadata and uploads to regulatory portals — **~600 cases/month, 15–20 min saved per case**.
- **AI Document Comparison & Validation Bot** (Python, VB.NET): compares report versions, extracts differences and auto-updates judgement sections — **~90% reduction in manual review effort**.

#### RPA Developer — Suzhou Transcosmos Information Creation Co., Ltd.

Jul 2020 – Jan 2021

Shanghai, China

- Designed and deployed unattended UiPath workflows (data extraction, web automation, email processing) in production, automating manual reporting and data entry with substantial time and error reductions; owned documentation, user training and bot reliability.

#### Research Assistant — Tongji University

Sep 2017 – Apr 2020

Shanghai, China

- Conducted theoretical and computational nuclear-physics research — designed and simulated nuclear-structure models using numerical methods and algorithmic optimisation. Authored **2 first-author peer-reviewed papers** (Nuclear Physics Review; Chinese Physics C). Tools: MATLAB, Fortran, Python.

## EDUCATION

**Master of Natural Science in Physics:** — Tongji University, Shanghai — 2017–2020

**Bachelor of Engineering in Thermal & Power Engineering:** — Shanghai Ocean University, Shanghai — 2007–2011

## PROJECTS (ALL OPEN-SOURCE · GITHUB.COM/STARS1210JASONHE)

### Meinrag — Multimodal RAG Platform

[github.com/stars1210JasonHe/Meinrag](https://github.com/stars1210JasonHe/Meinrag)

React 19 + FastAPI + FAISS/Chroma; hybrid retrieval (BM25 + dense, RRF) with LLM re-ranking; PDF bounding-box citations; Docling ingestion; Docker + PostgreSQL; test-covered. Powers the Cicero legal assistant in production.

### LAN Board-Game AI Platform

[github.com/stars1210JasonHe/lan-board-game-platform](https://github.com/stars1210JasonHe/lan-board-game-platform)

Gomoku / Chess / Xiangqi against LLM opponents or a friend over LAN. Custom Rust engines (alpha-beta + transposition tables) + minimax; Node / TypeScript / WebSocket; pluggable LLM providers (OpenAI / Anthropic / Ollama) + Stockfish.

### Meinopoly — Moddable Board-Game Engine

[github.com/stars1210JasonHe/Meinopoly](https://github.com/stars1210JasonHe/Meinopoly)

Config-driven, fully moddable Monopoly-style engine on boardgame.io; local/online multiplayer for 2–10 players; 296 automated tests (unit + Playwright E2E). In active development.

### Deutsch Learning Platform

[github.com/stars1210JasonHe/Deutsch\\_Learning\\_Platform](https://github.com/stars1210JasonHe/Deutsch_Learning_Platform)

FastAPI + Vue 3 German-learning app with a RAG guidance system (vector search + citations), OpenAI + DALL-E, JWT auth and Docker; 70K-word dictionary database.